

Divergent downstream biosynthetic pathways are supported by L-cysteine synthases of *Mycobacterium tuberculosis*

Reviewed Preprint

v2 • June 14, 2024

Revised by authors

Reviewed Preprint

v1 • December 4, 2023

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Abstract

Mycobacterium tuberculosis's (*Mtb*) autarkic lifestyle within the host involves rewiring its transcriptional networks to combat host-induced stresses. With the help of RNA-seq performed under various stress conditions, we identified that genes belonging to *Mtb* sulfur metabolism pathways are significantly upregulated during oxidative stress. Using an integrated approach of microbial genetics, transcriptomics, metabolomics, animal experiments, chemical inhibition, and rescue studies, we investigated the biological role of non-canonical L-cysteine synthases, CysM and CysK2. While transcriptome signatures of *RvΔcysM* and *RvΔcysK2* appear similar under regular growth conditions, we observed unique transcriptional signatures when subjected to oxidative stress. We followed pool size and labelling (^{34}S) of key downstream metabolites, viz. mycothiol and ergothioneine, to monitor L-cysteine biosynthesis and utilization. This revealed the significant role of distinct L-cysteine biosynthetic routes on redox stress and homeostasis. CysM and CysK2 independently facilitate *Mtb* survival by alleviating host-induced redox stress, suggesting they are not fully redundant during infection. With the help of genetic mutants and chemical inhibitors, we show that CysM and CysK2 serve as unique, attractive targets for adjunct therapy to combat mycobacterial infection.

eLife assessment

Sulphur atoms derived from cysteine are thought to play significant roles in maintaining redox homeostasis in *Mycobacterium tuberculosis*, which encounters stresses associated with immune cell interactions. In this **valuable** manuscript, the authors provide **solid** evidence that the genes encoding cysteine biosynthetic enzymes (cysM and cysK2) are required to maintain full viability of *M. tuberculosis* under in vitro stress conditions, macrophage infections, and within the lung tissues of mice. The manuscript presents transcriptomic and metabolomic evidence to support the hypothesis that CysM and CysK2 play distinct roles in maintaining cysteine-derived metabolite pools under stress conditions. The work will be of interest to microbiologists in general.

<https://doi.org/10.7554/eLife.91970.2.sa1>

Introduction

Mycobacterium tuberculosis (*Mtb*) continues to stride as the number one killer among all infectious diseases, accounting for nearly 1.5 million deaths yearly. The aggravating situation is despite the clinical use of over 20 antibiotics and a century-old vaccine, BCG. The gradual rise in the emergence of increasingly drug-resistant strains and HIV-TB co-infection further highlights the urgency to identify newer attractive drug targets. Throughout the course of infection, *Mtb* is exposed to a continuum of dynamic host-induced stresses such as severe nutrient deprivation, acidified compartments, and toxic reactive oxygen species (ROS) and reactive nitrogen species (RNS) produced by its resident phagosomes. In turn, *Mtb* produces copious amounts of actinomycetes-specific mycothiol, the major antioxidant in actinomycetes that act as the functional equivalent of glutathione, to combat ROS and RNS. In addition to mycothiol, *Mtb* also produces ergothioneine, a low molecular weight thiol, and several enzymes that act concertedly to subvert host-induced redox stress. The redox-active group of both mycothiol and ergothioneine is derived from L-cysteine. Hence, genes involved in the biosynthesis of L-cysteine are upregulated in the host and *in vitro* upon oxidative and nutritional stress [1–5]. Notably, an increased expression of these genes is functionally crucial, as suggested by the attenuated survival of transposon mutants of many sulfur and L-cysteine biosynthesis genes within the host [6]. In mycobacteria, sulfur assimilation begins with the import of sulfate through a sulfate transporter composed of SufI.CysT.W.A. Intracellular sulfate is a substrate for APS synthase CysD.N.C, which adenylates and phosphorylates sulfate to form adenosine 5'-phosphosulfate (APS) [7–9]. APS sits at a metabolic branch point; it can either be converted into sulfolipids [10] by consequent actions of multiple Stfs enzymes or reduced via SirA and SirH to sulfide (Figure S1). This pathway encompasses sulfide formation from sulfate, called the sulfur assimilation pathway [11, 12]. *Mtb* genome encodes three L-cysteine synthases – the canonical CysK1 and non-canonical CysM and CysK2 enzymes. Interestingly, humans do not possess L-cysteine synthases, raising the possibility of developing antibiotics without a homologous target in the host. CysK1 utilizes sulfide produced via the sulfur assimilation pathway and O-acetyl-L-serine produced from glycolytic intermediate 3-phosphoglycerate to produce L-cysteine [13–18]. CysM, on the other hand, uses O-phospho-L-serine and a small sulfur carrier protein CysO as substrates [15, 18]. Like CysK1, CysK2 utilizes O-phospho-L-serine and sulfide as substrates [19, 20] (Figure S1). In addition, *Mtb* can also synthesize L-cysteine through a reverse transsulfuration pathway from L-methionine. This example of convergent metabolic redundancy raises several interesting questions: (1) Why would *Mtb* rely on multiple enzymes and pathways to produce the same biomolecule? (2) Are these “functionally redundant” enzymes dispensable, or are they required at

a distinct cellular space, time, and condition? (3) Is the L-cysteine pool produced through a particular pathway functionally compartmentalized? That is, is it metabolized into a specific kind of downstream thiol?

To define these unsolved aspects of *Mtb* L-cysteine metabolism, we sought to investigate the interplay of non-canonical L-cysteine synthases of *Mtb* and elucidate their roles in abetting virulence. We aimed to decipher the relative contribution of CysM and CysK2 enzymes in alleviating host-induced stresses and promoting the survival of *Mtb* within the host. We also investigated their role in secondary metabolism, synthesizing low molecular weight thiols, such as mycothiol and ergothioneine, and understanding the consequential effects of their deletion on the global transcriptome of *Mtb*. Lastly, with the help of specific inhibitors, we evaluated their potential to serve as attractive drug targets for adjunct antibiotic therapy.

Results

Non-canonical L-cysteine synthases facilitate *Mtb* in combating host-induced stresses

Mtb is a generalist, a prototroph organism that can produce all 20 proteinogenic amino acids. In agreement with this notion, numerous microarray studies depict the upregulation of multiple amino acid pathways within the host [3, 4, 12], indicating a higher dependency of *Mtb* survival and virulence on amino acid biosynthesis and regulation. In response to infection, host immune responses often try to contain the bacillary growth by depriving the amino acid levels in intracellular environment. As a counter mechanism, *Mtb* has been shown to upregulate biosynthesis of amino acids such as tryptophan, lysine, and histidine to facilitate mycobacterial survival within the host [21, 22]. We sought to identify distinct host stresses that result in the transcriptional modulation of specific amino acid biosynthetic pathways with the help of RNA sequencing (RNA-Seq). We compared the transcriptional profile of *H37Rv* (*Rv*) grown in 7H9-ADC with the profiles obtained when bacilli were subjected to oxidative, nitrosative, starvation, and acidic stresses (Table S1). The volcano plot illustrates differentially expressed genes (DEGs) that were significantly upregulated (blue) and downregulated (red) under indicated stress conditions (absolute $\log_2$ Fold change > 1 and P_{adj} < 0.05) (Figure S2a-e). Exposure to starvation conditions resulted in drastic transcription modulation compared with other stresses, suggesting that nutrient deficiency is the primary driver of transcriptome remodelling. In contrast, we observed the lowest number of DEGs when the bacteria were subjected to mildly acidic conditions (pH 5.5). Heat maps of normalized DEGs depict that DEG changes were comparable across biological replicates within each sample set (Figure S2f-j). To better understand the RNA-seq results, we plotted the fold change of differentially expressed genes due to different stress conditions (Figure S3 & Table S2). This allowed us to understand the expression profile of genes in all the stress conditions simultaneously, regardless of whether they were identified as differentially expressed. The data revealed that specific clusters of genes are up- and downregulated in oxidative, SDS, and starvation conditions. In comparison, the differences observed in the pH 5.5 and nitrosative conditions were limited (Figure S3 & Table S2).

To further refine our understanding of the DEGs, we grouped them into various functional categories and found that genes belonging to intermediary metabolism & respiration remained the most affected in all conditions highlighting their role of metabolic rewiring (Figure S4a). Pathway enrichment analysis of the most enriched Gene Ontology (GO) further revealed that, while, expectedly, metabolic pathways were found to be downregulated during starvation, we observed enrichment of nitrogen metabolism (Figure S4b & Table S3). SDS stress resulted in the upregulation of branched-chain amino acids/keto acids degradation pathways (Figure S4c), and nitrosative stress ensued up-regulation of fatty acid and lipid biosynthetic processes (Figure S4d). There were no changes observed in mildly acidic conditions (Figure S4e). Importantly,

oxidative stress resulted in significant up-regulation of genes involved in sulfur metabolism (**Figure S4f**). We further analyzed the DEGs involved in sulfur and L-cysteine metabolism across sample sets and discerned an overlap of the genes affected under two or more conditions. Interestingly, we observed upregulation of sulfate transporters genes (*subI*, *cysT*, *cysW*, *cysA1*) across multiple stresses and sulfur assimilation (*cysH*, *cysA2*, *cysD*, *cysNC*) during starvation, oxidative and SDS stress. *CysK2*, a non-canonical L-cysteine synthase, was found to be up-regulated during all stresses except SDS stress (**Figure 1a**).

Thus RNA-Seq data suggest that genes involved in sulfur assimilation and L-cysteine biosynthetic pathway are upregulated during various host-like stresses in *Mtb* (**Figure S4**). Given the importance of sulphur metabolism genes in *in vivo* survival of *Mtb* [23, 24] it is not surprising that diverse environment cues dynamically regulate these genes. Microarray studies have shown upregulation of genes encoding sulphate transporter upon exposure to hydrogen peroxide and nutrient starvation [1, 4, 25–27]. Similarly, ATP sulfurylase and APS kinase are induced during macrophage infection and by nutrient depletion. Induction of these genes that coordinate the first few steps of the sulphur assimilation pathway indicates a probable increase in biosynthesis of sulphate-containing metabolites that may be crucial against host-inflicted stresses. Furthermore, genes involved in synthesis of reduced sulphur moieties (*cysH*, *sirA* and *cysM*) are also induced by hydrogen peroxide and nutrient starvation. Sulfur metabolism has been postulated to be important in transition to latency. This hypothesis is based on transcriptional upregulation of *cysD*, *cysNC*, *cysK2*, and *cysM* upon exposure to hypoxia. Multiple transcriptional profiling studies have reported upregulation of *moeZ*, *mec*, *cysO* and *cysM* genes when cells were subjected to oxidative and hypoxic stress [2, 18, 23, 26–29] further suggesting an increase in the biosynthesis of reduced metabolites such as cysteine and methionine and sulfur containing cell wall glycolipids upon exposure to oxidative stress. To address the functional relevance of this observation, we deleted two non-canonical L-cysteine synthases-CysM [5] and CysK2, from the *Mtb* chromosome. CysK2 mutant, *RvΔcysK2*, was generated using the recombineering method, and the recombination at the native loci was confirmed with the help of multiple PCRs (**Figure 1b**). While deletion of *cysM* or *cysK2* did not affect mycobacterial growth under *in vitro* nutrient-rich 7H9-ADC or 7H9-ADS (**Figure S5a** and **S5b**), it significantly compromised *Mtb* growth in defined Sauton's media ($\sim 2 \log_{10}$; **Figure 1c**), suggesting the importance of *cysM*- and *cysK2*- derived L-cysteine. Restoring *cysM* or *cysK2* expression in the complementation strains, *RvΔcysM::M* and *RvΔcysK2::K2*, rescued the growth defects (**Figure S5** & **Figure 1c**). *Mtb* is a metabolically versatile organism capable of utilizing a large variety of carbon and nitrogen sources [30]. Unlike 7H9, wherein glucose, glycerol, glutamate, and ammonia act as carbon and nitrogen sources, Sauton's media contains glycerol and asparagine as the sole carbon and nitrogen sources, suggesting metabolic reprogramming aided by CysM and CysK2 enable *Mtb* to grow optimally in a limited nutritional environment. This observation was further recapitulated under nutrient starvation (PBS), wherein the survival of *RvΔcysM* and *RvΔcysK2* was lower than parental *Rv* or *RvΔcysM::M* or *RvΔcysK2::K2* strains (**Figure 1d**). Interestingly, when exposed to acidic conditions (pH 4.5), *RvΔcysM* and *RvΔcysK2* survival were observed to be ~ 0.85 and $\sim 0.24 \log_{10}$ lower, respectively, compared with *Rv*, suggesting CysM is relatively more important for bacillary survival under acidic stress (**Figure 1e**). The highest attenuation of *RvΔcysM* and *RvΔcysK2* was observed upon the addition of cumene hydroperoxide (CHP), an organic hydroperoxide that, upon decomposition, generates free radicals. The relative survival of *RvΔcysM* and *RvΔcysK2* was ~ 2.04 and $\sim 1.31 \log_{10}$ lower, respectively, compared to *Rv* (**Figure 1f**). The addition of diamide, which results in thiol oxidation, also attenuated the survival of *RvΔcysM* and *RvΔcysK2* by ~ 1.33 and $\sim 1.26 \log_{10}$ compared with *Rv* (**Figure S5c**). When the mutant strains were subjected to nitrosative stress, the survival of *RvΔcysM* and *RvΔcysK2* was ~ 0.93 and $\sim 0.86 \log_{10}$ lower than *Rv* (**Figure 1g**). However, no significant attenuation was found during reductive and SDS stress (**Figure 1h-i**). Collectively, *RvΔcysM* and *RvΔcysK2* displayed increased susceptibility towards oxidative, nitrosative, mild acidification, and

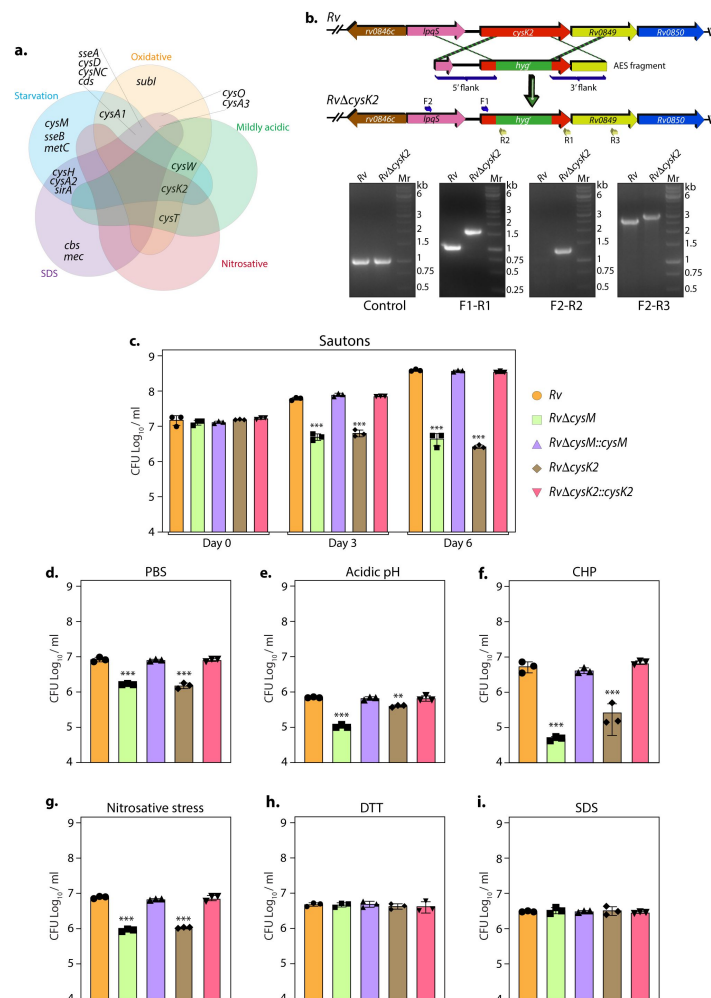


Fig 1

Non-canonical L-cysteine synthases facilitate mycobacterial survival upon host-like induced stresses *in vitro*.

(a) 5-way Venn diagram highlighting DEGs belonging to sulfur metabolism pathway under indicated stress conditions (absolute $\log_2$ Fold change ≥ 0.5 and P_{adj} value < 0.05). (b) Line Diagram illustrating the *cysK2* loci in *Rv* and *RvΔcysK2* and the strategy employed for the replacement of *cysK2* with *hyg^r*. Primer sets used for confirming the generation of *RvΔcysK2* are indicated as arrows. The first agarose gel image shows PCR amplicons with a control (*sigB*) gene-specific primer indicating the presence of nearly equivalent amount of genomic DNA isolated from *Rv* and *RvΔcysK2*. The second panel shows PCR amplicons with gene-specific primer set (F1-R1) in *Rv* and *RvΔcysK2* mutant, the third panel shows amplicons (F2-R2) expected only in *RvΔcysK2* mutant, and the fourth panel shows amplicons (F2-R3) in *Rv* and *RvΔcysK2* mutant confirming legitimate recombination at native loci. Mr represents 1-kb gene ruler ladder. (c-i) *Rv*, *RvΔcysM*, *RvΔcysK2*, *RvΔcysM::cysM*, and *RvΔcysK2::cysK2* strains were inoculated in Sauton's (c), PBS (d) or acidic (e), oxidative 50 μ M Cumene hydroperoxide (CHP) for 24 h (f), nitrosative (g), reductive (h) and SDS (i). Bar graphs represent the bacillary survival with data point indicating values CFU $\log_{10}$ /mL $\pm$ standard deviations (SD) from individual replicate (n=3). Statistical significance was drawn in comparison with *Rv* using one-way ANOVA followed by a *post hoc* test (Tukey test; GraphPad prism). ** $P < 0.005$; *** $P < 0.0005$.

PBS starvation to varying degrees compared with *Rv*, *RvΔcysM::M* and *RvΔcysK2::K2*. The data suggests that L-cysteine, produced via CysM and CysK2, and its downstream products help mycobacteria thwart specific stresses that *Mtb* encounters within the host.

Distinct roles of CysM and CysK2

To decipher the mechanism through which CysM and CysK2 combat oxidative stress, we performed a global transcriptomic analysis of *Rv*, *RvΔcysM*, and *RvΔcysK2* in the presence and absence of oxidative stress (CHP) (Table S4). Principal Component Analysis demonstrated clear separation of strains under different conditions. Intriguingly, while *RvΔcysM* and *RvΔcysK2* were closely located on a PCA plot in the absence of any stress, oxidative stress resulted in a significant divergence between the two groups (**Figure 2a** [↗](#)). Deletion of *cysM* resulted in differential expression of 322 genes (159 downregulated and 163 upregulated) (**Figure S6a-b** [↗](#)), while deletion of *cysK2* impacted 278 genes (155 downregulated and 123 upregulated) under regular growth conditions (**Figure S6c-d** [↗](#)) (absolute log₂ Fold change>1 and Padj<0.05). In contrast, upon treatment with CHP, nearly ~33% and ~53% of *Mtb* genes were differentially expressed in *RvΔcysM* and *RvΔcysK2*, respectively, compared with *Rv* (**Figure 2b-c** [↗](#)). To understand the individual contribution of CysM and CysK2 in combating oxidative stress, we compared the DEGs of *RvΔcysK2* to *RvΔcysM* under regular growth and oxidative stress conditions. While DEGs between *RvΔcysK2* and *RvΔcysM* were limited to 13 in untreated conditions (**Figure 2d** [↗](#)), CHP treatment resulted in differential expression of 1372 genes (**Figure 2e** [↗](#) and **Figure S6e** [↗](#)), highlighting unique transcriptional signatures associated with each L-cysteine synthase under oxidative stress. Subsequently, we compared the DEGs of *Rv*, *RvΔcysM*, and *RvΔcysK2* obtained under oxidative stress with the help of a Venn diagram. This analysis revealed that CysM and CysK2 influence a shared repertoire of 1023 genes (529 upregulated and 494 downregulated) compared to *Rv* upon CHP treatment (**Figure 2f** [↗](#)). Importantly, CysM and CysK2 distinctively modulate 324 and 1104 genes, respectively, during oxidative stress (**Figure 2f-g** [↗](#)). In addition to being directly regulated by the synthases, it is highly likely that the changes in the expression of these genes is because of distinct downstream consequences. Assortment of DEGs into functional classes revealed intermediary metabolism & respiration and cell wall pathways among the most impacted categories (**Figure 2h** [↗](#)). These results corroborate the recent finding that CysK2 alters the phospholipid profile of the *Mtb* cell envelope [31 [↗](#)]. Pathway enrichment analysis of the most enriched Gene ontology biological process indicated that while most pathways are commonly affected by CysM and CysK2, these L-cysteine synthases also have a unique transcriptional footprint (Table S5). Pathways such as DNA binding, homologous recombination, translation, ribosome, etc., were commonly affected in *RvΔcysM* and *RvΔcysK2* compared to *Rv* during oxidative stress (**Figure S6f, g** [↗](#)). Genes belonging to oxidative phosphorylation and quinone binding were upregulated, while DNA binding and cholesterol catabolism gene categories were found to be downregulated in *RvΔcysM* compared to *RvΔcysK2* during oxidative conditions (**Figure 2i** [↗](#)). Together, our data point to the unique roles of CysM and CysK2, as their deletion differentially affects various cellular pathways under oxidative stress.

Key metabolites are differentially affected upon CysM and CysK2 deletion

L-Cysteine concentration in *Mtb* is notoriously low, usually 10-fold lower than most other amino acids [32 [↗](#)]. As expected, L-cysteine levels were below the limit of detection/quantification (not shown). Therefore, we used key downstream metabolites to monitor L-cysteine biosynthesis and utilization. We followed pool size and labelling (Na³⁴SO₄) of L-methionine, mycothiol/mycothione, and ergothioneine, made from L-cysteine, via three distinct and dedicated metabolic pathways (**Figure S7** [↗](#)). First, L-methionine, mycothiol, and ergothioneine concentrations vary from zero to 24 h, indicating that their pool sizes are not as stable as those of other core metabolites. This is consistent with their function, susceptibility to redox homeostasis, and L-methionine's role in protein synthesis. Second, upon challenge with CHP for 24 h, L-methionine levels decrease in *Rv*,

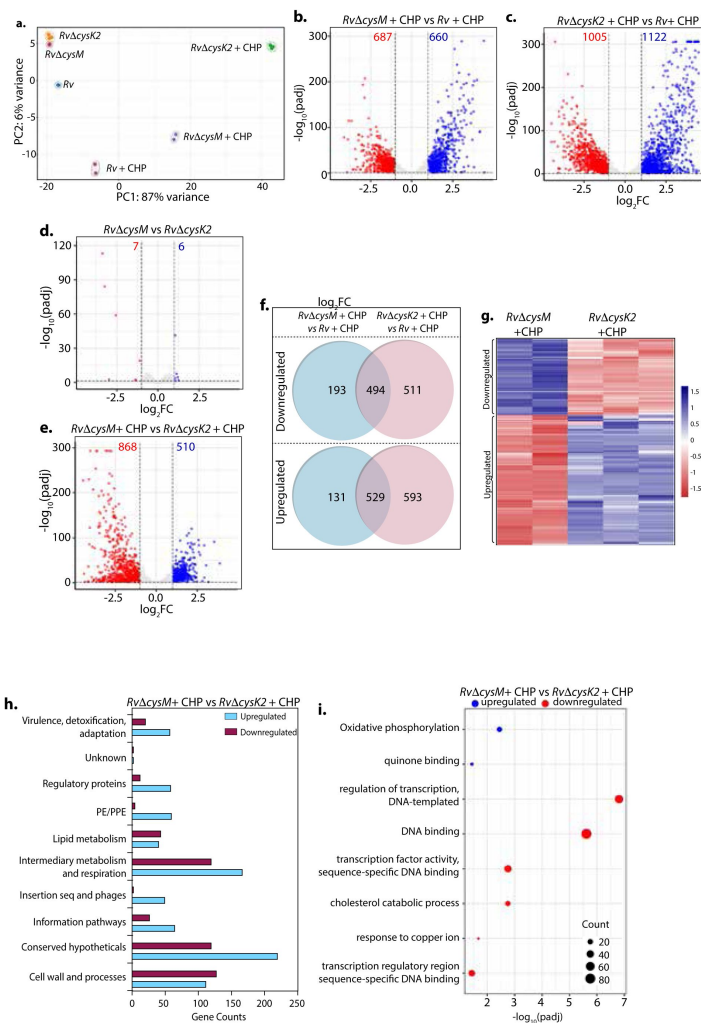


Fig 2

Distinct roles of CysM and CysK2 in attuning cellular processes.

(a) PCA plot demonstrating separation of various bacterial strains under different conditions (b-e) Volcano plots illustrating significantly upregulated (blue) and downregulated (red) genes in indicated strains and conditions with absolute $\log_2$ Fold change ≥ 1 and P_{adj} value < 0.05 . Numbers in the top quadrant highlight the number of significantly upregulated (blue) and downregulated (red) genes in each condition (f) Venn diagram showing the number of significantly downregulated and upregulated DEGs that overlap between indicated strains (g) Heat Maps depicting normalized gene count of differentially expressed genes (DEGs) in independent replicates of *RvΔcysM* and *RvΔcysK2* grown under oxidative conditions with absolute $\log_2$ Fold change ≥ 1 and P_{adj} value < 0.05 . The colour intensity indicates relative upregulated (blue) and downregulated (red) genes compared to the control. (h) Horizontal bar graph depicting the number of DEGs belonging to a particular functional category upon oxidative stress in *RvΔcysM* compared to *RvΔcysK2* under oxidative conditions. (i) Pathway enrichment by DAVID depicting significantly enriched Gene Ontology (GO) biological processes based on DEGs upon oxidative stress in *RvΔcysM* compared to *RvΔcysK2* (absolute $\log_2$ Fold change > 1 and $P_{adj} < 0.05$).

remain constant in *RvDcysK2*, and increase in *RvDcysM* (Figure 3a). Ergothioneine levels are identical in the three strains, and upon treatment with CHP, they are decreased in the parent strain but remain unchanged in the *RvDcysK2* and *RvDcysM* mutant (Figure 3b). Mycothiol levels increase in the *Rv*, *RvDcysK2* strains and remain nearly constant in *RvDcysM* (Figure 3c). Finally, mycothione levels increased slightly in the parent strain, decreased in the *RvDcysK2*, and remained stable in the *RvDcysM* strain (Figure 3d). These changes revealed a significant role of distinct L-cysteine biosynthetic routes on redox stress and homeostasis.

Employing ^{34}S labelling, we observed a reduced rate of synthesis of mycothiol, mycothione and ergothioneine in *RvΔcysM* compared to *Rv* and *RvΔcysK2* (Figure 3e-g). Interestingly, ergothioneine levels were found to be reduced in both *RvΔcysM* and *RvΔcysK2* compared to *Rv*, possibly underlying one of the reasons for their enhanced sensitivity to oxidative stress.

This result further indicates that these two routes are partially redundant; that is, they generate the same end product, L-cysteine. Yet, pool size measurements demonstrate significant changes, highlighting that while these pathways produce the same metabolite, their complex biological roles and requirements (co-substrates, pathways, regulation, etc.) are not fully redundant. This interpretation is in accordance with transcriptomics and phenotypic results described above. These non-overlapping metabolic requirements (e.g., O-acetyl-L-serine vs. O-phospho-L-serine vs. CysM) are likely the source of the different metabolic phenotypes observed. Therefore, even producing the same end-product, genetic disruption of different L-cysteine synthases differently affects bacterium metabolism and fitness, leading to the distinct phenotypes observed between *RvDcysK2* and *RvDcysM* *in vitro*, *in cellulo*, and *in vivo*.

CysM and CysK2 alleviate the toxicity of host-produced ROS and RNS

To define whether the attenuation of mutant strains observed during defined *in vitro* host-like conditions can be recapitulated within the host cells, we compared the survival of parental, mutants, and complementation strains in murine peritoneal macrophages (Figure 4a). Survival of *RvΔcysM* and *RvΔcysK2* was ~ 1.22 and $\sim 0.85 \log_{10}$ lower, respectively, compared with *Rv* at 96 h post-infection (p.i) (Figure 4b). Notably, the addition of L-cysteine alleviated survival differences (Figure 4c), suggesting that reduced L-cysteine levels in mutant strains are responsible for their attenuated survival. To further validate that CysM and CysK2 support *Mtb* survival by detoxifying peroxides generated by the host, we compared the survival of strains in the peritoneal macrophages extracted from wild-type C57BL/6 (WT) mice that are proficient in eliciting oxidative and nitrosative stress or murine strains that are deficient in their ability to produce hydrogen peroxide (H_2O_2) (*phox^{-/-}*) or both oxidative and nitrosative stress (*IFN γ ^{-/-}*). As anticipated, regardless of the mice genotype, there was no difference in the survival of *Rv* and complementation strains (Figure 4d-e). On the other hand, attenuation of *RvΔcysM* and *RvΔcysK2* observed in wild-type macrophages is partially reversed in the macrophages obtained from *phox^{-/-}* mice (Figure 4d) and completely nullified in macrophages isolated from *IFN γ ^{-/-}* mice (Figure 4e). To further dissect the relative contribution of CysM and CysK2 in combating ROS and RNS independently or collectively, the survival of *Mtb* strains was examined in peritoneal macrophages isolated from WT and *phox^{-/-}* in the presence or absence of iNOS inhibitor. While the inhibition of ROS or RNS independently only partially alleviated the attenuated survival *RvΔcysM* and *RvΔcysK2*, inhibition of both ROS and RNS by treating peritoneal macrophages isolated from *phox^{-/-}* with iNOS inhibitor completely nullified survival defects of the mutants (Figure 4f). These results suggest that non-canonical L-cysteine synthases, CysM and CysK2, play an essential role in combating host-induced redox stress.

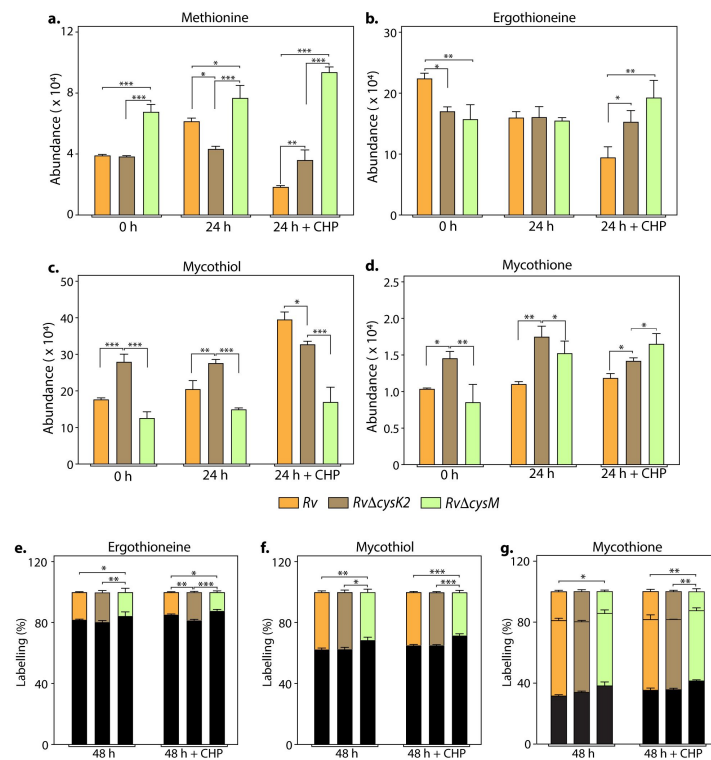


Fig 3

Key metabolites are differentially affected upon CysM and CysK2 deletion

(a-d) Total abundance (Ion count/Protein concentration) of L-methionine (a), ergothioneine (b), mycothiol (c), and mycothione (d) in *Rv* with or without 50 μ M CHP treatment. (e-g) The percentage of labelled (colored) and unlabeled (black) ergothioneine (e), mycothiol (f), and mycothione (g) in *Rv* with or without 50 μ M CHP treatment at 48 h. Percentage was calculated with respect to 0 h abundance for each replicate. Bars depict the mean of biological replicates (n=3), and error bars represent the standard deviation. The same samples were used for analysis (a-d) and (e-g). Statistical significance was calculated between *Rv* and *RvΔcysM*/*RvΔcysK2*; and between *RvΔcysM* and *RvΔcysK2* using non-parametric t-test. * $P < 0.05$; ** $P < 0.005$; *** $P < 0.0005$.

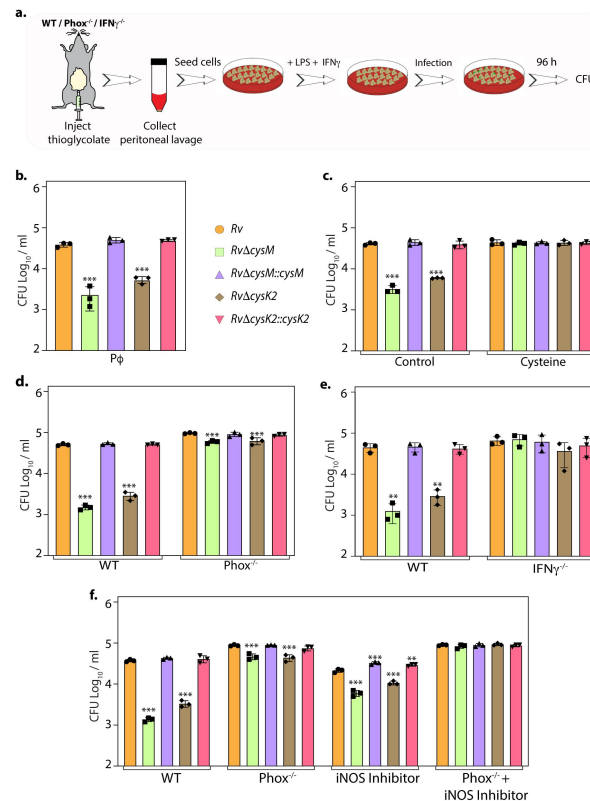


Fig 4

L-cysteine synthases ameliorate mycobacterial survival in response to host-induced oxidative and nitrosative stress

(a) Pictorial representation of peritoneal macrophage infection experiments. Thioglycolate was injected into the peritoneum cavity of C57BL/6, *phox*^{-/-}, *IFN γ* ^{-/-} mice. Four days post-injection, peritoneal macrophages were extracted and activated by treatment with IFN γ overnight and LPS - 2 h prior to the infection. In specific cases, iNOS inhibitor 1400 W was added along with IFN γ /LPS. The intracellular bacillary survival was calculated 96 h p.i. **(b)** Peritoneal macrophages from C57BL/6 mice were infected, and intracellular bacillary survival was enumerated 96 h p.i. **(c)** Peritoneal macrophages from C57BL/6 mice were infected, left untreated (control) or treated with 0.2mM L-cysteine at the time of infection and intracellular bacillary survival was assessed 96 h p.i. **(d-f)** Peritoneal macrophages isolated from mice of indicated genotypes were either left untreated or treated with iNOS inhibitor, 1400W. Intracellular bacillary survival was enumerated 96 h p.i. **(b-f)** Data points are presented as CFU log₁₀/ml $\pm$ SD of each replicate (n=3). Statistical significance was drawn in comparison with *Rv* using one-way ANOVA followed by a *post hoc* test (Tukey test; GraphPad prism). ***P* < 0.005; ****P* < 0.0005.

CysM and CysK2-derived L-cysteine support mycobacterial survival *in vivo*

Given the importance of non-canonical L-cysteine synthases in mitigating host-induced redox stress, we next sought to examine whether CysM and CysK2 independently facilitate mycobacterial survival in a murine infection model. We followed disease progression by enumerating colony forming units (CFU) of the lung and spleen of the infected mice at day 1, 4, and 8 weeks p.i (Figure 5a & d). CFUs enumerated on day 1 showed equal bacillary deposition across strains in the lungs (Figure 5b & e). Compared with *Rv*, survival of *RvΔcysM* was $\sim 1.25 \log_{10}$ lower at 4 weeks p.i, which further attenuated to $\sim 1.71 \log_{10}$ at 8 weeks p.i in the lungs (Figure 5b). Similarly, dissemination of *RvΔcysM* in the spleen was $\sim 1.63 \log_{10}$ lower at 4 weeks p.i and $\sim 1.72 \log_{10}$ lower at 8 weeks p.i compared with *Rv* (Figure 5c). Similarly, *RvΔcysK2* was $\sim 1.67 \log_{10}$ and $\sim 1.87 \log_{10}$ lower at 4 and 8 weeks p.i, respectively, compared with *Rv* in the lungs (Figure 5e) and $\sim 1.63 \log_{10}$ and $\sim 1.72 \log_{10}$ lower at 4 and 8 weeks p.i, respectively in the spleens of infected mice (Figure 5f).

To understand whether the ability of CysM to mitigate oxidative and nitrosative stress is linked to its role in facilitating mycobacterial survival *in vivo*, we infected WT, *phox*^{-/-} and *IFNγ*^{-/-} mice with *Rv*, *RvΔcysM* and *RvΔcysM::M* and analyzed bacillary load at 4 weeks p.i (Figure 5g). As shown in Figures 5h & 5i, the survival of *RvΔcysM* is impaired in WT mice that produce both ROS and RNS, compared with *phox*^{-/-} and *IFNγ*^{-/-} mice. The survival defect observed due to the absence of CysM in the lungs was partially rescued in *phox*^{-/-} mice. Due to the lack of ROS and RNS in *IFNγ*^{-/-} mice, *RvΔcysM* showed higher bacillary load than in *phox*^{-/-} mice (Figure 5h). *RvΔcysM* displayed better survival in *phox*^{-/-} and *IFNγ*^{-/-} mice in the spleen than lungs, which could either be because of relatively lesser ROS and RNS stress or imperiled response in clearing *Mtb* infection (Figure 5i). In contrast to the peritoneal macrophage infection experiment (Figure 4), attenuated survival of *RvΔcysM* was not completely salvaged in *IFNγ*^{-/-} mice, suggesting that CysM may also be involved in alleviating additional stresses such as nutrient deprivation and/or *IFNγ* independent ROS/RNS produced by the host. Data presented suggests that L-cysteine produced through non-canonical pathways are independently important for mycobacterial survival *in vivo*.

L-Cysteine synthase inhibitors can effectively kill *Mtb* within the host

Data presented above demonstrated that CysM and CysK2 might serve as clinically-important targets for adjunct TB therapy. Brunner *et al.* screened a compound library to identify inhibitors of mycobacterial L-cysteine synthases-CysK1, CysK2, and CysM. We selected three compounds - Compound 1 (C1) (named compound 2 in [28]), which inhibits all three synthases; Compound 2 (C2) (compound 6 in [28]), which inhibits both non-canonical L-cysteine synthases and Compound 3 (C3) (compound 31 in [28]), which selectively inhibits CysK1 (Figure 6a & b). To examine the therapeutic potential of these inhibitors, we first tested their effect on the survival of *Rv* within host peritoneal macrophages. Treatment with C1 resulted in $\sim 1 \log_{10}$ attenuation compared with the untreated cells. Killing mediated by C2 or C3 was marginally lower compared with C1 (Figure 6c). Next, we examined the ability of these drugs to enhance the bactericidal activity of INH, a drug whose activity depends on redox state of the cell [33], and therefore could be affected by disruption of L-cysteine downstream pathways. Towards this goal, we first measured the MICs of INH (0.06 μg/ml) and C1 (0.6 mg/ml), C2 (0.6 mg/ml), and C3 (0.15 mg/ml) independently. As expected, all the compounds were ineffective in killing *Mtb* because of redundant roles and relatively less requirement of L-cysteine synthases during regular growth conditions. To examine the combinatorial effect, we selected sub-MIC values of INH (0.03 μg/mL, encircled orange) and increasingly two-fold diluted concentrations of C1, C2, or C3, all below MIC values (Figure 6d-f; blue encircled denotes the starting concentration) and assessed bacterial

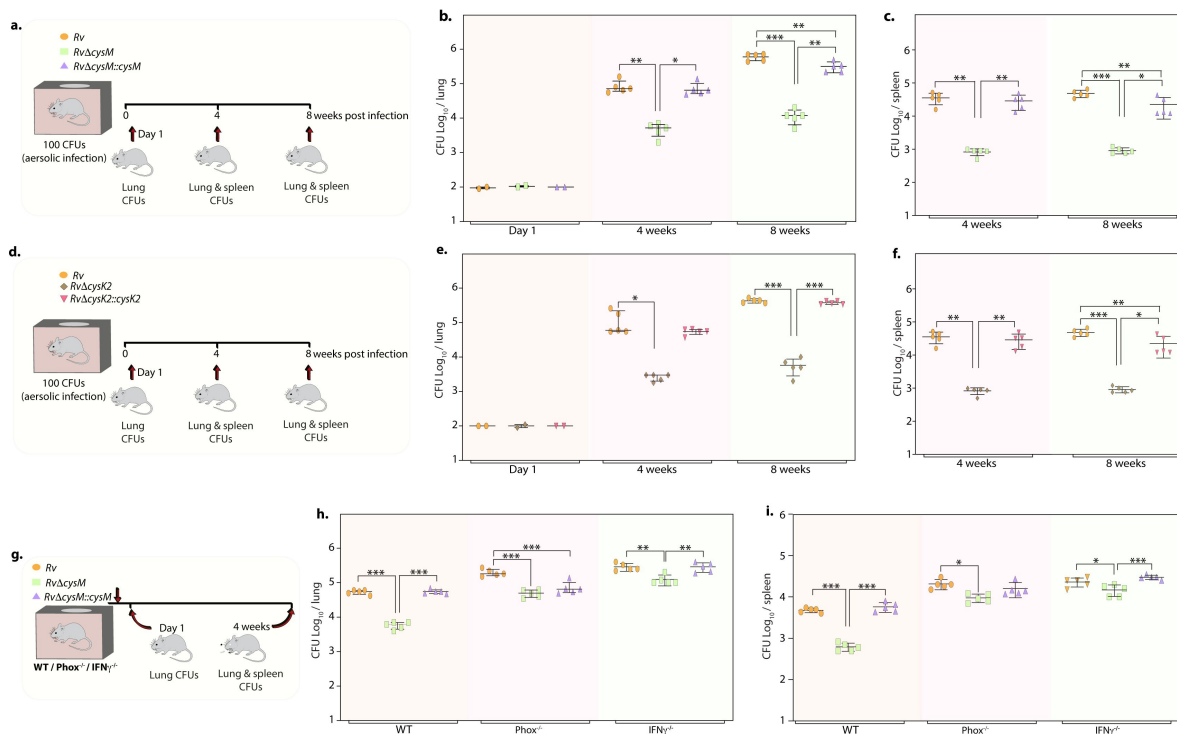


Fig 5

Deletion of non-canonical L-cysteine synthases attenuates mycobacterial survival in murine lungs and spleen

(a) Schematic outline of murine infection experiment. C57BL/6 (n = 12 per group) were infected with *Rv*, *RvΔcysM*, or *RvΔcysM::cysM* strains via an aerosol route. CFU was enumerated at day 1 (n=2), week 4 (n=5) and week 8 (n=5). **(b-c)** Each data point represents Log₁₀ CFU in lung (b) and spleen (c) of an infected animal, and the error bar indicates the median with interquartile range for each group. **(d)** Schematic outline of murine infection experiment. C57BL/6 (n = 12 per group) were infected with *Rv*, *RvΔcysK2*, or *RvΔcysK2::cysK2* strains via an aerosol route. CFU was enumerated at day 1 (n=2), week 4 (n=5) and week 8 (n=5). **(e-f)** Each data point represents Log₁₀ CFU in the lung (e) and spleen (f) of an infected animal, and the error bar indicates the median with interquartile range for each group. **(g)** Schematic outline of murine infection experiment. C57BL/6 (n = 7 per group), *phox*^{-/-} (n = 7 per group), *IFNγ*^{-/-} (n = 7 per group) were infected with *Rv*, *RvΔcysM*, or *RvΔcysM::cysM* strains via an aerosol route. CFU was enumerated at day 1 (n=2) and week 4 (n=5). **(h-i)** Each data point represents Log₁₀ CFU in the lung (h) and spleen (i) of an infected animal, and the error bar indicates the median with interquartile range for each group. **(a-i)** Statistical significance was drawn in comparison with *Rv* using one-way ANOVA followed by a *post hoc* test (Tukey test; GraphPad prism). ****P* < 0.0005.

survival upon combinatorial treatment with INH and the three L-cysteine synthase inhibitors. The addition of either of the inhibitors below MIC rendered *Mtb* highly susceptible to INH, highlighting the potency of these compounds to serve as an adjunct therapy to combat mycobacterial infection.

Discussion

Throughout its course of infection, *Mtb* must persist in a hostile, oxidizing, and nutrient-deprived environment of host macrophage. Understanding the dynamic metabolic interactions between the pathogen and its host is imperative to identify its weaknesses which can be exploited to design new-age chemotherapy. Our study provides convincing evidence that the genes involved in the L-cysteine biosynthetic pathway are attractive targets for the design of anti-mycobacterial drugs. Genes involved in sulfur assimilation and L-cysteine biosynthesis were found to be upregulated in the transcriptome profile of *Rv* subjected to various host-like stresses. The sulfur metabolism pathway was particularly enriched upon the addition of oxidizing agent, CHP (**Figure 1**, S2 & S3). This observation has been consistently reported by multiple studies demonstrating the up-regulation of sulfur assimilation and L-cysteine biosynthetic genes in response to oxidative stress, nutrient deprivation, and macrophage infection [1–5]. Besides being involved in protein synthesis, L-cysteine is also important for the biosynthesis of L-methionine, S-adenosyl methionine, coenzyme A, and iron-sulfur clusters. As a thiol-containing molecule, L-cysteine contributes to the intracellular redox state directly and through the production of major antioxidants like mycothiol and ergothioneine. These numerous essential functions of L-cysteine prompted us to further examine its roles in the context of *Mtb* cellular metabolism and virulence.

Unlike *Mtb*Δ*cysH*, which was reported to be an L-cysteine auxotroph and thus required L-methionine or glutathione supplementation to grow *in vitro* (from which L-cysteine can be generated catabolically) [24], neither deletion of *cysK2* nor *cysM* impacted *Mtb* growth kinetics *in vitro* suggesting that these genes are functionally redundant during rich growth conditions [5] [31] (**Figure S5**). Interestingly, the induction of various host-like stresses attenuated the growth of these mutants compared to the parental strain, pointing towards the possibility of an enhanced requirement of L-cysteine-derived antioxidants and other biomolecules to subvert these stresses (**Figure 1**). In agreement with this hypothesis, we previously showed that the cellular thiol levels are higher during oxidative stress in *Mtb* [5]. Similarly, various MSH and ERG mutants display enhanced sensitivity to oxidative stress caused by treatment with H₂O₂, cumene hydroperoxide, or O₂^{•−} [34–38].

While the transcriptomic profiles of mutants were highly similar to each other during regular growth conditions, the addition of CHP resulted in differential expression of >30% *Mtb* genes, indicating that cues and downstream effects of the two non-canonical L-cysteine synthases are partially non-overlapping, further underlying their importance for the bacillary growth inside the host (**Figure 2**). We also found that the steady-state levels (pool sizes) of key L-cysteine derived antioxidants, ergothioneine, and mycothiol, are significantly different at 24 h (**Figure 3**). The rate of synthesis of mycothiol, mycothione and ergothioneine were observed to be lower in *Rv*Δ*cysM* compared to *Rv* and *Rv*Δ*cysK2*.

The addition of L-cysteine in oxidatively stressed cells nullified the compromised survival of the mutants indicating that *Mtb* cells are able to uptake L-cysteine from the extracellular medium, as shown previously [5], and reduced levels of L-cysteine in the mutants are chiefly responsible for their attenuated growth in the presence of stresses (**Figure 4**). Various *Mtb* mutants in mycobacterial sulfur metabolism genes are severely compromised to persist within the host and cause disease [24, 39–44]. Inorganic sulfate is present at 300–500 μM in human plasma [44]. However, the inability of host-derived sulfur/L-cysteine to compensate for attenuated survival of mutants is suggestive of either inaccessibility of sufficient L-cysteine in *Mtb* niches or inefficient expression, function, or uptake by transporters *in vivo*. Importantly, compromised

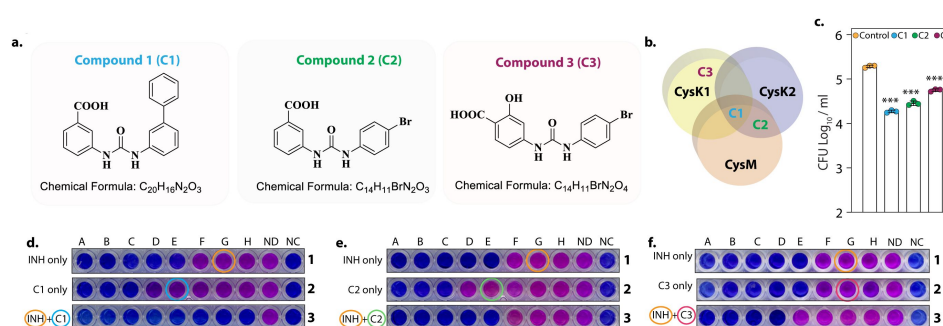


Fig 6

L-cysteine synthases inhibitors can effectively kill Mtb within the host.

(a) Chemical structure and formula of lead compounds. **(b).** Venn diagram illustrating the specificity of compounds on mycobacterial L-cysteine synthases. **(c).** Peritoneal macrophages were infected with Rv. 4 h post-infection, cells were either left untreated (control) or 5 mg/ml C1 (a), C2 (b) or C3 (c) was added to the infected cells. Bar graphs represent mean Log_{10} CFU/ml $\pm$ SD, and values from independent replicates are represented by individual data points (n=3). **(d-f).** Alamar blue assay is used to determine MIC values of C1 (d), C2, and C3 (f) with and without Isoniazid in *Rv*. Starting concentration of INH in 1A is 0.96 $\mu\text{g/ml}$, 1B-0.43 $\mu\text{g/ml}$ and each subsequent column 2 fold dilution. Concentration of C1, C2 and C3 were 2.4 mg/ml in 2A and 1.2 mg/ml in 2B and in each subsequent column 2 fold dilution. The concentration of INH + C1 or C2 in 3A was 0.015 $\mu\text{g/ml}$ + 0.15 mg/ml, and 3B was 0.0075 $\mu\text{g/ml}$ + 0.075 mg/ml. The concentration of INH + C3 in 3A was 0.015 $\mu\text{g/ml}$ + 37 $\mu\text{g/ml}$, and 3B was 0.0075 $\mu\text{g/ml}$ + 18.5 and each subsequent column 2-fold dilution. ND-no drug and NC-no culture.

survival of *RvΔcysM* and *RvΔcysK2* *in vivo* was largely mitigated in IFN γ ^{-/-} and Phox^{-/-}, indicating that the non-canonical L-cysteine synthases, CysM and CysK2, independently facilitate mycobacterial survival during immune-mediated redox stress (**Figure 5** [↗](#)).

Using network analysis of the *Mtb* protein interactome, a flux balance analysis of the reactome and randomized transposon mutagenesis data, along with sequence analyses and a structural assessment of targetability, Raman *et al.*, reported CysK2 and CysM as high confidence drug targets [\[45\]](#) [↗](#). Similarly, inhibitors of CysM showed mycobactericidal activity in a nutrient-starvation model of dormancy [\[46\]](#) [↗](#). Interestingly, humans do not reduce sulfur to produce L-cysteine; they rather synthesize L-cysteine through SAM-dependent transmethylation followed by transsulfuration of L-methionine. Owing to their complete absence in humans, mycobacterial L-cysteine biosynthetic genes and their regulators represent unique, attractive targets for therapeutic intervention [\[47\]](#) [↗](#). We found that both *RvΔcysM* and *RvΔcysK2* displayed enhanced antibiotic sensitivity *in vitro* and within the host (data not shown). Importantly, a combination of L-cysteine synthase inhibitors with front-line TB drugs like INH, significantly reduced the bacterial survival *in vitro* (**Figure 6** [↗](#)). Altogether, this study demonstrates for the first time that the two non-canonical L-cysteine synthases have non-redundant biological functions. Deletion or biochemical inhibition of CysM or CysK2 perturbs redox homeostasis of *Mtb* and allow for the maximal effect of host macrophages antibacterial response, and thus increased elimination of virulent *Mtb*.

Methods

Bacterial strains and culturing conditions

Mtb culturing conditions were performed as described previously [\[5\]](#) [↗](#), [48\]](#) [↗](#).

Generation of *RvΔcysK2* mutant and complementation strain

We generated gene replacement mutants through the recombineering method [\[49\]](#) [↗](#) as previously described [\[5\]](#) [↗](#), [48\]](#) [↗](#). Briefly, 671bp upstream of 176th nucleotide from 5' end (5' flank) and 643 bp downstream of 943rd nucleotide from 3' end (3' flank) of *cysK2* were PCR amplified from *Rv* genomic DNA using Phusion DNA polymerase (Thermo Scientific). The amplicons were digested with BstAPI and ligated with compatible hygromycin resistance (*hyg*^r) cassette and *oriE* +cos λ fragments [\[49\]](#) [↗](#), to generate the allelic exchange substrate (AES). AES was digested with SnaBI to release the LHS-*hyg*^r-RHS fragment, and the eluted fragment was electroporated into the recombineering proficient *Rv* -ET strain [\[5\]](#) [↗](#). Multiple *hyg*^r colonies were examined by PCRs to screen for legitimate recombination at *cysK2* locus. *RvΔcysK2*, thus generated, was cured of pNit-ET through negative selection on LB Agar plates containing 2% sucrose. To generate the complementation strain, full length *cysK2* gene was PCR amplified from genomic DNA isolated from *Rv* as the template and Phusion DNA polymerase (ThermoFischer Scientific). The amplicon was digested with NdeI-HindIII (NEB), cloned into the corresponding sites in pNit-3F and the resultant plasmid was electroporated into *RvΔcysK2* to generate the *RvΔcysK2::cysK2* strain. *RvΔcysM* and *RvΔcysM::cysM* strains were generated in the lab previously [\[5\]](#) [↗](#) using a similar method [\[2\]](#) [↗](#).

Ex vivo infection experiments

Balb/c, C57BL/6 (B6), phox^{-/-} (B6.129S6-Cybbtm1Din/J; JAX# 002365) or IFN- γ ^{-/-} (B6.129S7-Ifngtm1Ts/J; JAX#002287) mice were procured from The Jackson Laboratory. 4% thioglycolate (Hi-Media) was injected into the peritoneum cavity of 4- to 6-weeks old mice and peritoneal macrophages were extracted four days post-injection and seeded and processed as described [\[5\]](#) [↗](#). In specific cases, the peritoneal macrophages were treated with 10 ng/ml IFN γ (BD Biosciences) overnight and 10 ng/ml lipopolysaccharide (LPS, Sigma) for 2 h for activation. Where

indicated, cells were further pretreated with 100 μ M 1400 W (Sigma) overnight to inhibit iNOS or pretreated with 0.2 mM L-cysteine before infection. Single-cell *Mtb* suspensions were used for infection at 1:10 (host cells: bacteria) MOI. 4 h post-infection (p.i), cells were washed thrice and replenished with complete RPMI containing IFN γ + LPS, 1400 W, or L-cysteine, as required. The infected host cells were washed thrice with PBS, lysed using 0.05% Dodecyl sulfate sodium salt (SDS), and serial dilutions were plated on 7H11-OADC to enumerate bacillary survival.

In vivo infection experiments

Mice (4–6-weeks) housed in ventilated cages at the Tuberculosis Aerosol Challenge Facility at the International Centre for Genetic Engineering and Biotechnology (New Delhi, India) were infected via aerosol route with $\sim$ 100 bacilli using the Madison Aerosol Chamber (University of Wisconsin, Madison, WI). At 24 h post-infection, mice (n=2; per group) were euthanized to determine the bacterial deposition. At 4/8 weeks p.i, lungs and spleen were homogenized and plated on 7H11+OADC containing PANTA.

RNA isolation and qRT-PCRs

Mtb strains were cultured in triplicates in 7H9-ADS till O.D. reached 0.3-0.4. One set was left untreated (control), the other was treated with 50 μ M of CHP for 6 h. A culture equivalent to 10 O.D.₆₀₀ was resuspended in TRIzol (Invitrogen). Zirconium beads were added to the cell-Trizol mix to facilitate lysing *Mtb* cells with the help of a bead beater (MP FastPrep system, MP Biomedicals) and RNA was extracted and analyzed as described [54]. Data was plotted as $2^{(-\Delta\Delta C_t)}$ wherein the gene expression was normalized with respect to 16s rRNA (*rrs* gene), followed by normalization with control strain/condition/group.

RNA sequencing and analysis

Total RNA was isolated from two-three biological replicates of indicated *Mtb* strains, and their concentrations were checked by Qubit (ThermoFischer Scientific) followed by quality assessment through Agilent 2100 BioAnalyzer (Agilent RNA 6000 Nano Kit). Both sets of RNA seq represented in **Figures 1** and **5** were processed and run at the same time, the same set of control (*Rv*) triplicates were used for the analysis of both figures. Samples with RIN values $\geq$ 7 were processed for RNA sequencing using the Illumina NovaSeq 6000 Platform (CSIR-CCMB central facility; read length of 100 bp, 20 million paired-end reads / sample).

Illumina adapters and low-quality reads were discarded and those with quality scores $<$ 20 and smaller than 36 bp were eliminated from raw sequencing reads using cutadapt [50]. Processed reads were mapped to the *Mtb* H37Rv, (https://ftp.ncbi.nlm.nih.gov/genomes/refseq/bacteria/Mycobacterium_tuberculosis/reference/GCF_000195955.2_ASM19595v2/), using hisat2 with default parameters [51]. Uniquely aligned reads were counted with the help of feature Counts of Subread package [52] and those with total read count $<$ 10 across all the samples were removed, and the rest were used for further downstream analysis. Differentially expressed genes (DEGs) were identified using DESeq2 [53] and those with adjusted p-value $<$ 0.05 and absolute log₂ Fold change $>$ 1 or 0.5 were considered. The raw read counts were rlog normalized the raw read counts for PCA plot and heat map with the DESeq2 package.

Functional enrichment analysis

Functional enrichment analysis was performed with DAVID web services [54]. We specifically used GO terms and KEGG pathways for this analysis. Only top 10 enriched hits based on gene counts were plotted.

Preparation of samples for metabolomics

Mtb strains were grown in 7H9 media until an OD 1 and then inoculated onto 0.22µm nitrocellulose filters and grown on 7H10 plates containing 0.5g/L BSA Fraction V, 0.2% dextrose and 0.085% NaCl (ADS) for 5 days. The filters were then transferred to 7H10 plus ADS plates containing sodium sulfate-³⁴S (Merck 718882) containing either 50µM CHP or no CHP for 24 and 48 h. The metabolites were extracted by mechanical lysis in cold acetonitrile/methanol/water (2:2:1) containing 0.1mm acid washed Zirconia beads. The lysates were clarified by centrifugation and filtered through a 0.22µm Spin-X column (Costar). The lysates were mixed 1:1 with acidified acetonitrile (0.1% formic acid). Due to the tendency of *M. tuberculosis* to form clumps, which significantly skew any cell number estimations we normalized samples to protein/peptide concentration using the BCA assay kit (Thermo). Therefore, our LC-MS data is express as ion counts/mg protein or ratios of that for the same metabolite. This is a standard way to express ion abundance data [10 [↗](#), 12 [↗](#)].

Liquid chromatography – mass spectrometry metabolomics

LC–MS analysis was done in an Agilent 1290 Infinity II HPLC connected to a 6230B time-of-flight (ToF) mass spectrometer using a Dual AJS ESI ionization source. Compounds were separated in a Cogent Diamond Hydride Type C silica column (2.1 × 150 mm). Solvent A was LC–MS grade H₂O + 0.1% (v/v) formic acid and solvent B was acetonitrile + 0.1% (v/v) formic acid. The gradient was from 85% B to 5% B over 14 min. Flow rate was 0.4 mL/min. The ion source parameters were as follows: Gas temperature 250 °C, Drying gas flow rate 13 l/min, nebulizer pressure 35 psig, sheath gas temperature 350 °C, sheath gas flow 12 l/min, capillary voltage 3500 V and nozzle voltage 2000 V, The ion optic voltages were 110 V for the fragmentor, 65 V for the skimmer and 750 for the octopole radio frequency voltage. MS data were analyzed with the MassHunter suite version B0.7.0.00.

MIC analysis

MIC values were assessed using Alamar Blue assay, as described previously [55 [↗](#)]. Briefly, 100µl 7H9-ADS medium without Tween 80 was added to each well of a 96-well plate. First well of each column were filled with 100µl of the test drug/antibiotic, which was then serially diluted across the column. *Mtb* cells corresponding to 0.01 A₆₀₀ were diluted in 100µl 7H9-ADS medium and were added to each well. Two rows one in which the drug was not added (no drug [ND]), and the other wherein *Mtb* cells were replaced with 7H9-ADS (no cells [NC]) acted as controls. The 96 well plate was sealed with parafilm and kept at 37°C. After 5 days, 20µl of 0.25% filter-sterilized resazurin was added to each well, and color development was captured after 24 h.

Statistical analysis

Unless otherwise specified, experiments were performed in triplicates and repeated independently at least twice. CFU results were plotted, and significance of the datasets was calculated using one-way ANOVA followed by a *post hoc* test (Tukey test) on GraphPad Prism 5. Figures were customized using Adobe Illustrator version 26.3.1. Statistical significance was set at *P* values < 0.05 significant (*, *P* < 0.05; **, *P* < 0.005; ***, *P* < 0.0005). Source datasets can be requested from the corresponding author.

Data availability

RNA Seq data is available at the NCBI Gene Expression Omnibus Database, accession no. GEO GSE225792, the link to the database is <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE225792> [↗](#) and the reviewer token for private access is ipmnekaifmlbcj.

Acknowledgements

We thank Dr. Apruva Sarin for *phox*^{-/-} mice and thoughtful discussion; ICGEB for access to their TACF; CCMB for access to their sequencing facility. MZK was supported by Research Associateship from SERB (CRG/2018/001294). Research reported in this publication was supported by the core grant of the National Institute of Immunology; DBT grant BT/PR13522/COE/34/27/2015; SERB grant CRG/2018/001294, and JC Bose fellowship JCB/2019/000015 to VKN. Work in LPSC's laboratory was supported by the Francis Crick Institute, which receives its core funding from Cancer Research UK Grant FC001060, UK MRC Grant FC001060, and Wellcome Trust Grant FC001060.

Author contributions

MZK, LPSC and VKN conceptualized the study; MZK contributed to generation of constructs and strains; BS and MZK were involved in the preparation of RNA sequencing samples, while NKS and DTS performed the analyses. MZK and BS performed animal studies at ICGEB; MZK contributed to *in vitro* stress susceptibility assays, macrophage assays and inhibitor studies; DMH and LPSC performed the metabolomic studies; DVSP and DS synthesized chemical inhibitors; MZK curated the data. MZK, YK, LPSC and VKN wrote the manuscript; VKN and LPSC acquired funding for the study. All authors read and approved the manuscript.

Ethics statement

Protocols for animal experiments were preapproved by the Animal Ethics Committee of the National Institute of Immunology, New Delhi, India (IAEC numbers 409/16 and 462/18) as per standard institutional guidelines.

Table S1: Differentially Expressed Genes (DEG) analysis *Rv* under stress vs *Rv*.

Table S2: We compiled a list of differentially expressed genes in at least one of the five comparisons against control.

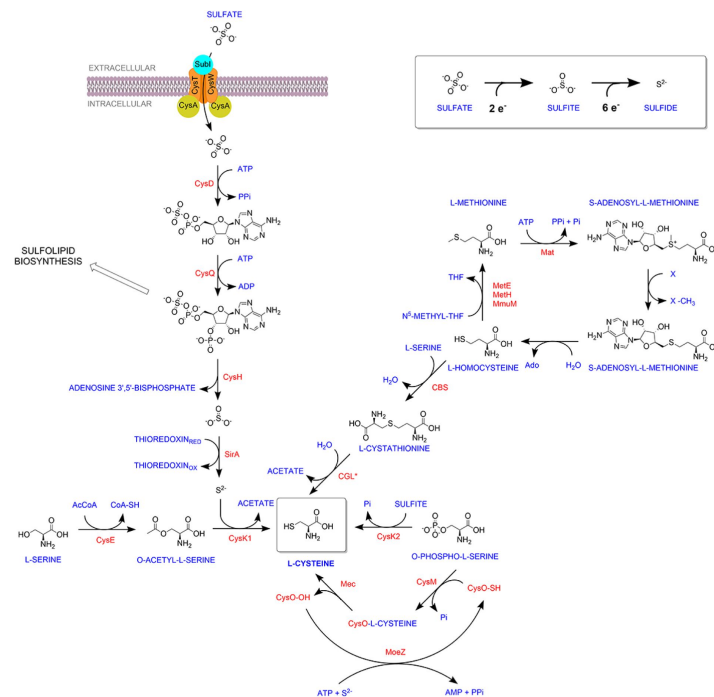
Table S3: Pathway enrichment analysis of DEGs from Table S1.

Table S3: DEG analysis of *RvΔcysM* or *RvΔcysK2* vs *Rv* with or without CHP stress & DEG analysis *RvΔcysM* +CHP vs *RvΔcysK2* + CHP.

Table S4: Pathway enrichment analysis of DEGs from Table S3.

References

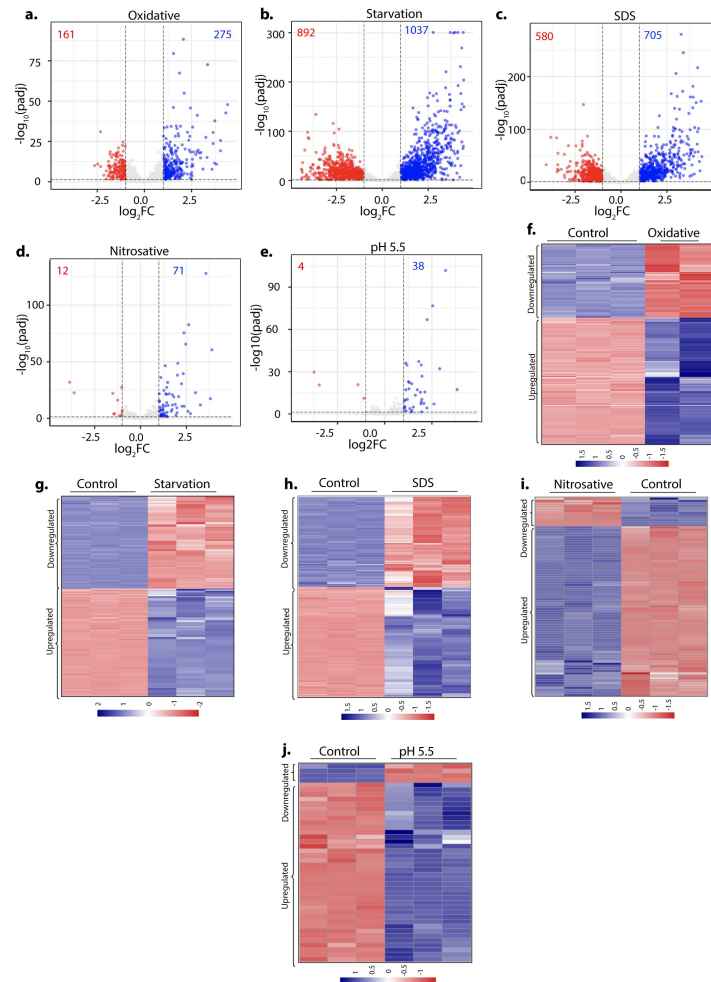
1. Hampshire T., et al. (2004) **Stationary phase gene expression of *Mycobacterium tuberculosis* following a progressive nutrient depletion: a model for persistent organisms?** *Tuberculosis (Edinb)* **84**:228–38
2. Manganelli R., et al. (2002) **Role of the extracytoplasmic-function sigma factor sigma(H) in *Mycobacterium tuberculosis* global gene expression** *Mol Microbiol* **45**:365–74



Supplementary Figure 1

Overview of Sulfur metabolism pathway in *Mycobacterium tuberculosis*

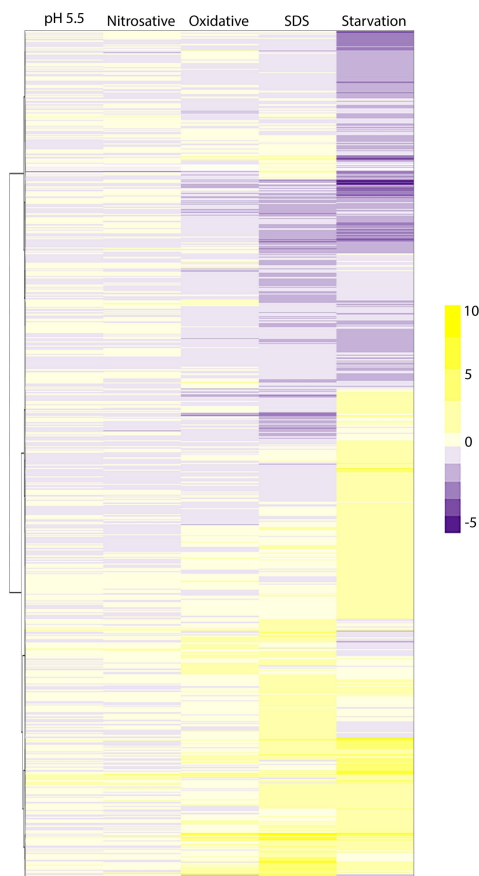
Sulfate is imported via the sulfate transporter composed of SufI.CysT.W.A, which is acted upon by CysDNC complex to form Adenosine-5'-phosphosulfate (APS). APS can either form sulfolipids or is reduced by subsequent actions of CysH and SirA to form sulfide. CysK1, the classical L-cysteine synthase, utilizes reduced sulfide and O-acetyl-L-serine as substrates to form L-Cysteine. Alternatively, L-cysteine is produced by CysM, a non-canonical L-cysteine synthase that is unique to actinomycetes and uses O-phospho-L-serine and a small sulfur carrier protein CysO to form L-cysteine. *Mtb* also encodes for a third L-cysteine synthase, CysK2. However, the major product of this pathway is S-sulfo-L-cysteine, which can get rapidly converted to L-cysteine. The reverse transsulfuration pathway is also depicted.



Supplementary Figure 2

Sulfur assimilation and L-cysteine biosynthesis pathways are upregulated under stress conditions.

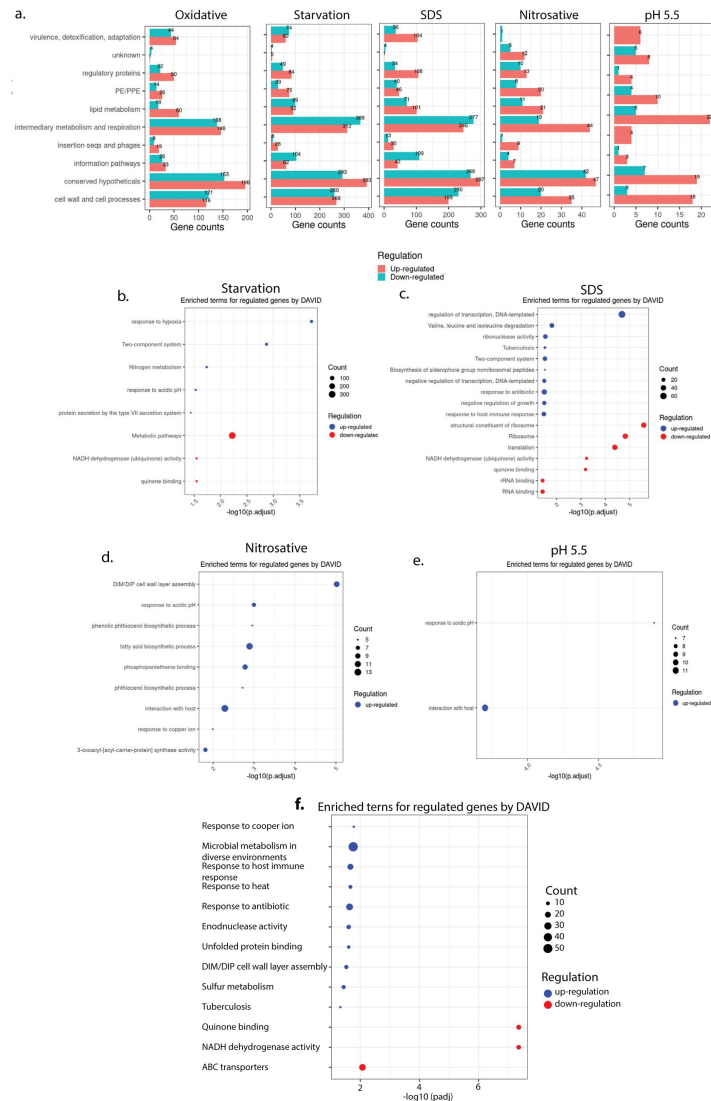
(a-e) Volcano plots illustrating significantly upregulated (blue) and downregulated (red) genes in *Rv* grown under indicated conditions with Log_2 Fold change ≥ 1 and P_{adj} value < 0.05 compared to control (untreated) *Rv*. Numbers in the top quadrant depict the number of significantly upregulated (blue) and downregulated (red) genes in each condition compared to the control. (f-j) Heat Maps depicting normalized gene count of differentially expressed genes (DEGs) in independent replicates of *Rv* grown under indicated conditions with Log_2 Fold change ≥ 1 and P_{adj} value < 0.05 . The colour intensity indicates relative upregulated (blue) and downregulated (red) genes compared to the control.



Supplementary Figure 3

Genes that were differentially expressed in at least one of the five stresses.

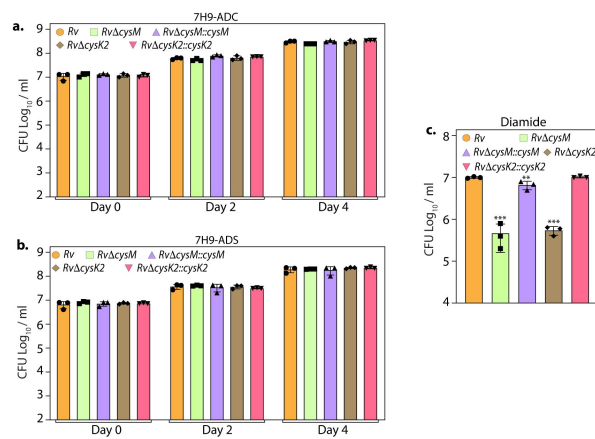
We compiled a list of genes differentially expressed in at least one of the five stresses against compared untreated *Rv*. We made a data frame of fold changes for these selected genes in the five comparisons. We then generated a heatmap where the five comparisons were plotted as columns (x-axis), and selected genes were plotted as rows (y-axis). Hierarchical clustering was performed on the y-axis so that genes with similar fold changes across comparisons clustered together.



Supplementary Figure 4

Genes belonging to the Sulfur metabolism pathway are upregulated upon oxidative stress

(a) Horizontal bar graph demonstrating the number of DEGs belonging to a particular functional category under indicated stress condition. **(b-e)** Pathway enrichment by DAVID depicting significantly enriched Gene Ontology (GO) biological processes based on DEGs in starvation (b), SDS (c), Nitrosative (d) and pH 5.5 (e). **(f)** Pathway enrichment by DAVID depicting significantly affected Gene Ontology (GO) biological processes based on DEGs (Log_2 Fold change ≥ 0.5 and P_{adj} value < 0.05).



Supplementary Figure 5

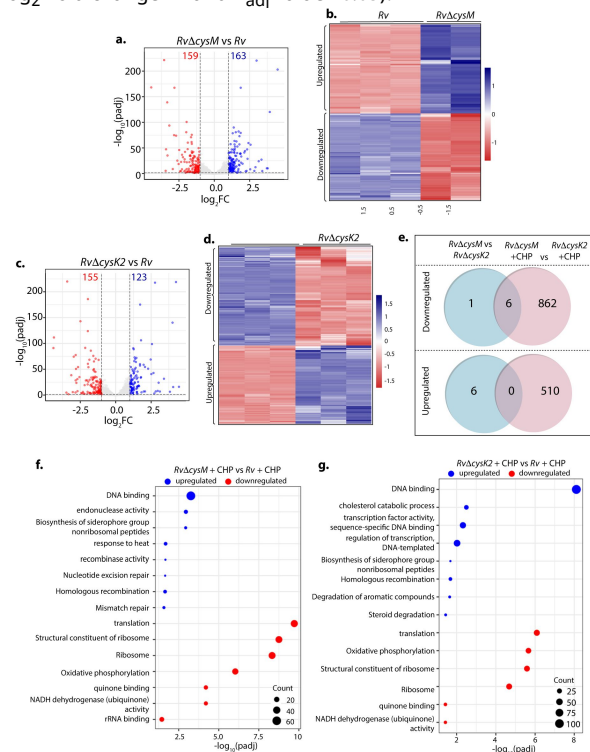
L-cysteine synthases facilitate mycobacterial survival upon oxidative stresses *in vitro*

(a-c) *Rv*, *RvΔcysM*, *RvΔcysK2*, *RvΔcysM::cysM*, and *RvΔcysK2::cysK2* strains were inoculated in 7H9-ADC (a), 7H9-ADS (b), 7H9-ADS plus 10mM diamide (c) medium. Bar graphs represent the bacillary survival with data point indicating values CFU log₁₀/mL ± SD from individual replicate (n=3). Statistical significance was drawn in comparison with *Rv* using one-way ANOVA followed by a *post hoc* test (Tukey test; GraphPad prism). ***P* < 0.005; ****P* < 0.0005.

Supplementary Figure 6

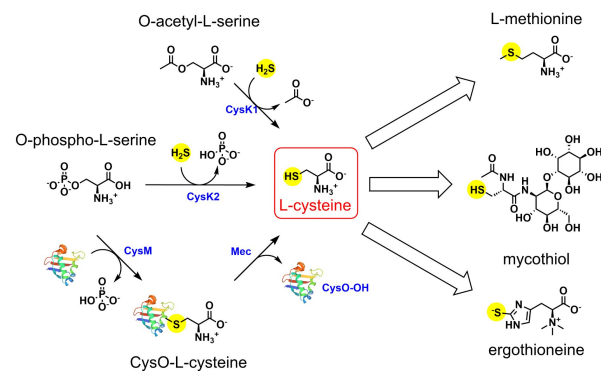
Unique transcriptional signatures of CysM and CysK2.

(a & c) Volcano plots illustrating significantly upregulated (blue) and downregulated (red) genes in indicated strains and conditions with $\log_2$ Fold change > 1 and P_{adj} value < 0.05 . Numbers in the top quadrant highlight the number of significantly upregulated (blue) and downregulated (red) genes in each condition (b & d) Heat Maps depicting normalized gene count of differentially expressed genes (DEGs) in independent replicates of *RvΔcysM* and *RvΔcysK2* compared to *Rv* with $\log_2$ Fold change > 1 and P_{adj} value < 0.05 . The colour intensity indicates relative upregulated (blue) and downregulated (red) genes compared to the control. (e) Venn diagram showing the number of significantly downregulated and upregulated DEGs that overlap between indicated strains ($\log_2$ Fold change > 1 and P_{adj} value < 0.05). (f & g) Pathway enrichment by DAVID depicting significantly enriched Gene Ontology (GO) biological processes based on DEGs upon oxidative stress in *RvΔcysM* (f) and *RvΔcysK2* (g) compared to *Rv* ($\log_2$ Fold change > 1 and P_{adj} value < 0.05).



Supplementary Figure 7

Overview of metabolites tracked through ^{35}S sulfur incorporation



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Reviewer #1 (Public Review):

Summary:

Khan et. al., investigated the functional redundancy of the non-canonical L-cysteine synthases of *M. tuberculosis*, CysM and CysK2, focussing on their role in mitigating the effects of host-derived stress. They found that while deletion mutants of the two synthases (RvΔcysM, RvΔcysK2) have similar transcriptomes under standard conditions, their transcriptional response to oxidative stress is distinct. The impact of deleting the synthases also differentially affected the pools of L-cysteine-derived metabolites. They show that the mutants (RvΔcysM, RvΔcysK2) have impaired survival in peritoneal macrophages and in a mouse model of infection. Importantly, they show that the survival of the mutants increases when the host are defective in producing reactive oxygen and nitrogen species, linking the phenotype to a defect in combating host-derived stress. Finally, they show that compounds inhibiting L-cysteine synthases reduces intracellular survival of *M. tuberculosis*.

Strengths:

(1) The distinct transcriptome of the RvΔcysM and RvΔcysK2 mutants in the presence of oxidative stress provides solid evidence that these mutants are distinct in their response to oxidative stress, and suggests that they are not functionally redundant.

(2) The use of macrophages from *phox*^{-/-} and *INF*^{-/-} mice and an iNOS inhibitor for the intracellular survival assays provides solid evidence that the survival defect seen for the RvΔcysM and RvΔcysK2 mutants is related to their reduced ability to combat host-derive oxidative and nitrosative stress. This is further supported by the infection studies in *phox*^{-/-} and *INF*^{-/-} mice.

Weaknesses:

Inclusion of the complemented strains in the metabolite study would strengthen the data. Furthermore, using an alternate method to quantify the MSH:MSSM ratio would provide

insight into the redox homeostasis in mutants in the presence and absence of CHP to support the statement that "deletion or inhibition of CysM or CysK2 perturbs redox homeostasis of Mtb".

The authors sought to investigate the functional redundancy of the non-canonical L-cysteine synthases CysM and CysK2. While their distinct transcriptional response to oxidative stress suggests distinct physiological roles, the study did not explore these differences, and therefore provides only preliminary insight into the underlying reasons for this observation. In the context of drug development, this work suggests that while L-cysteine synthase inhibitors do not have high potency for killing intracellular *M. tuberculosis*, they have potential for decreasing the pathogen's survival in the presence of host-derived stress.

<https://doi.org/10.7554/eLife.91970.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Khan et al., investigated the functional redundancy of the non-canonical L-cysteine synthases of M. tuberculosis, CysM and CysK2, focussing on their role in mitigating the effects of host-derived stress. They found that while deletion mutants of the two synthases (RvΔcysM, RvΔcysK2) have similar transcriptomes under standard conditions, their transcriptional response to oxidative stress is distinct. The impact of deleting the synthases also differentially affected the pools of L-cysteine-derived metabolites. They show that the mutants (RvΔcysM, RvΔcysK2) have impaired survival in peritoneal macrophages and in a mouse model of infection. Importantly, they show that the survival of the mutants increases when the host is defective in producing reactive oxygen and nitrogen species, linking the phenotype to a defect in combating host-derived stress. Finally, they show that compounds inhibiting L-cysteine synthases reduce the intracellular survival of M.

tuberculosis.

Strengths:

(1) The distinct transcriptome of the RvΔcysM and RvΔcysK2 mutants in the presence of oxidative stress provides solid evidence that these mutants are distinct in their response to oxidative stress, and suggests that they are not functionally redundant.

(2) The use of macrophages from phox-/- and INF-/- mice and an iNOS inhibitor for the intracellular survival assays provides solid evidence that the survival defect seen for the RvΔcysM and RvΔcysK2 mutants is related to their reduced ability to combat host-derived oxidative and nitrosative stress. This is further supported by the infection studies in phox-/- and INF-/- mice.

Weaknesses:

(1) There are several previous studies looking at the transcriptional response of M. tuberculosis to host-derived stress, however, the authors do not discuss initial RNA-seq data in the context of these studies. Furthermore, while several of the genes in sulfur assimilation and L-cysteine biosynthetic pathway genes are upregulated by more than

one stress condition, the data does not support the statement that it is the "most commonly upregulated pathway in Mtb exposed to multiple host-like stresses".

We have made changes in the manuscript in line with reviewer's suggestion.

"Thus RNA-Seq data suggest that genes involved in sulfur assimilation and L-cysteine biosynthetic pathway are upregulated during various host-like stresses in Mtb (Figure S2). Given the importance of sulphur metabolism genes in in vivo survival of Mtb [1, 2], it is not surprising that these genes are dynamically regulated by diverse environment cues. Microarray studies have shown upregulation of genes encoding sulphate transporter upon exposure to hydrogen peroxide and nutrient starvation [3-7]. Similarly, ATP sulfurlyase and APS kinase is induced during macrophage infection and by nutrient depletion. Induction of these genes that coordinate first few steps of sulphur assimilation pathway indicate that probable increase in biosynthesis of sulphate containing metabolites that may be crucial against host inflicted stresses. Furthermore, genes involved in synthesis of reduced sulphur moieties (cysH, sirA and cysM) are also induced by hydrogen peroxide and nutrient starvation. Sulfur metabolism has been postulated to be important in transition to latency. This hypothesis is based on transcriptional upregulation of cysD, cysNC, cysK2, and cysM upon exposure to hypoxia. Multiple transcriptional profiling studies have reported upregulation of moeZ, mec, cysO and cysM genes when cells were subjected to oxidative and hypoxic stress [1, 6-11] further suggesting an increase in the biosynthesis of reduced metabolites such as cysteine and methionine and sulfur containing cell wall glycolipids upon exposure to oxidative stress [12]. We have modified the sentence to "significantly upregulated pathway in Mtb exposed to multiple host-like stresses"

(2) For the quantification of the metabolites, it isn't clear how the abundance was calculated (e.g., were standards for each metabolite used? How was abundance normalised between samples?), and this information should be included to strengthen the data.

Thanks for picking up this. We have extended our description of metabolomics methods. It now reads: "Due to the tendency of M. tuberculosis to form clumps, which significantly skews any cell number estimation we normalized samples to protein/peptide concentration using the BCA assay kit (Thermo). Therefore, our LC-MS data is expressed as ion counts/mg protein or ratios of that for the same metabolite. This is a standard way to express ion abundance data as it was done previously [13, 14].

Furthermore, labelling with L-methionine was performed to determine the rate of synthesis of the L-cysteine-derived metabolites. L-cysteine is produced from L-methionine via the transsulfuration pathway, which is independent of CysM and CysK2. It is therefore difficult to interpret this experiment, as the impact of deleting CysM and CysK2 on the transsulfuration pathway is likely indirect.

The reviewer may have misunderstood the experiment and the results presented. Labelling was not performed with L-methionine. We use ³⁴S derived from SO₄²⁻, to monitor reductive assimilation of sulfur and its transit from S₂⁻ until L-methionine, passing through cysteine. We specified in material and methods that we have used sodium sulfate-³⁴S (Merck 718882), as our label source of sulfur. This method was first employed in M. tuberculosis by the Bertozzi group to identify sulfolipids in mycobacteria. Therefore, we are not measuring transsulfuration, but instead direct synthesis of L-methionine via cysteine, and consequently we are indeed assessing the importance of cysK2 and cysM in this process. We have now added to the results section (page 9) that we employed (Na³⁴SO₄) for labeling, to make sure other readers will not think we are measuring transsulfuration.

(3) The ability of L-cysteine to rescue the survival defect of the RvΔcysM and RvΔcysK2 mutants in macrophages is interpreted as exogenous L-cysteine being able to compensate for reduced intracellular levels. However, there is no evidence that L-cysteine is being taken up by the mutants and an alternate explanation is that L-cysteine functions as an antioxidant within cells i.e., it reduces intracellular ROS.

The concentration of L-cysteine used for peritoneal macrophage survival rescue experiments was titrated to have no minimum survival advantage in case of wild-type Rv. Thus, at the given concentration, we believe that the contribution of cysteine in reducing intracellular ROS within cells does not have a major role since there is no significant difference in the survival of wild-type Rv strain. Had cysteine reduced intracellular ROS, we would expect increased bacterial survival of Rv due to diminished oxidative stress.

Furthermore, L-cysteine addition also mitigates CHP induced survival defect in vitro [15] and nullifies observed effect of Cysteine inhibitors in vitro [16] suggesting that cysteine or cystine can be transported into Mtb. This has also been previously shown in case of AoxR mutant strain [15], CysH [2] and over 70% uptake of exogenously added [35S] cysteine to a growing culture of Mtb [17].

The authors sought to investigate the functional redundancy of the non-canonical L-cysteine synthases CysM and CysK2. While their distinct transcriptional response to oxidative stress suggests distinct physiological roles, the study did not explore these differences and therefore provides only preliminary insight into the underlying reasons for this observation. In the context of drug development, this work suggests that while L-cysteine synthase inhibitors do not have high potency for killing intracellular M. tuberculosis, they have the potential to decrease the pathogen's survival in the presence of host-derived stress.

Reviewer #2 (Public Review):

Summary:

The paper examines the role L-cysteine metabolism plays in the biology of Mycobacterium tuberculosis. The authors have preliminary data showing that Mycobacterium tuberculosis has two unique pathways to synthesize cysteine. The data showing new compounds that act synergistically with INH is very interesting.

Strengths:

RNAseq data is interesting and important.

Weaknesses:

The paper would be strengthened if the authors were to add further detail to their genetic manipulations.

The authors provide evidence that they have successfully made a cysK2 mutant by recombineering. This data looks promising, but I do not see evidence for the cysM deletion. It is also important to state what sort of complementation was done (multicopy plasmid, integration proficient vector, or repair of the deletion). Since these mutants are the basis for most of the additional studies, these details are essential. It is important to include complementation in mouse studies as unexpected loss of PDIM could have occurred.

The details of CysM knockout generation have been previously published ([15]; Appendix Figure S4), and complementation strain details are provided in the methods section.

Reviewer #3 (Public Review):

In this work, the authors conduct transcriptional profiling experiments with Mtb under various different stress conditions (oxidative, nitrosative, low pH, starvation, and SDS). The Mtb transcriptional responses to these stress conditions are not particularly new, having been reported extensively in the literature over the past ~20 years in various forms. A common theme from the current work is that L-cysteine synthesis genes are seemingly up-regulated by many stresses. Thus, the authors focused on deleting two of the three L-cysteine synthesis genes (cysM and cysK2) in Mtb to better understand the roles of these genes in Mtb physiology.

The cysM and cysK2 mutants display fitness defects in various media (Sautons media, starvation, oxidative and nitrosative stress) noted by CFU reductions. Transcriptional profiling studies with the cysM and cysK2 mutants revealed that divergent gene signatures are generated in each of these strains under oxidative stress, suggesting that cysM and cysK2 have non-redundant roles in Mtb's oxidative stress response which likely reflects the different substrates used by these enzymes, CysO-L-cysteine and O-phospho-L-serine, respectively. Note that these studies lack genetic complementation and are thus not rigorously controlled for the engineered deletion mutations.

The authors quantify the levels of sulfur-containing metabolites (methionine, ergothioneine, mycothiol, mycothionine) produced by the mutants following exposure to oxidative stress. Both the cysM or cysK2 mutants produce more methionine, ergothioneine, and mycothionine relative to WT under oxidative stress. Both mutants produce less mycothiol relative to WT under the same condition. These studies lack genetic complementation and thus, do not rigorously control for the engineered mutations.

Next, the mutants were evaluated in infection models to reveal fitness defects associated with oxidative and nitrosative stress in the cysM or cysK2 mutants. In LPS/IFN γ activated peritoneal macrophages, the cysM or cysK2 mutants display marked fitness defects which can be rescued with exogenous cysteine added to the cell culture media. Peritoneal macrophages lacking the NADPH oxidase (Phox) or IFN γ fail to produce fitness phenotypes in the cysM or cysK2 mutants suggesting that oxidative stress is responsible for the phenotypes. Similarly, chemical inhibition of iNOS partly abrogated the fitness defect of the cysM or cysK2 mutants. Similar studies were conducted in mice lacking IFN γ and Phox establishing that cysM or cysK2 mutants have fitness defects in vivo that are dependent on oxidative and nitrosative stress.

Lastly, the authors use small molecule compounds to inhibit cysteine synthases. It is demonstrated that the compounds display inhibition of Mtb growth in 7H9 ADC media. No evidence is provided to demonstrate that these compounds are specifically inhibiting the cysteine synthases via "on-target inhibition" in the whole Mtb cells. Additionally, it is wrongly stated in the discussion that "combinations of L-cys synthase inhibitors with front-line TB drugs like INH, significantly reduced the bacterial load inside the host". This statement suggests that the INH + cysteine synthase inhibitor combinations reduce Mtb loads within a host in an infection assay. No data is presented to support this statement.

We agree with the reviewer that the experiments do not conclusively prove that these compounds specifically inhibit the cysteine synthases via "on-target inhibition" in the whole Mtb cells. However, the inhibitors used in this study have been previously profiled in vitro (<https://www.sciencedirect.com/science/article/abs/pii/S0960894X17308405?via%3Dihub>). We have modified the sentence to "a combination of L-cysteine synthase inhibitors with front-line TB drugs like INH, significantly reduced the bacterial survival in vitro"

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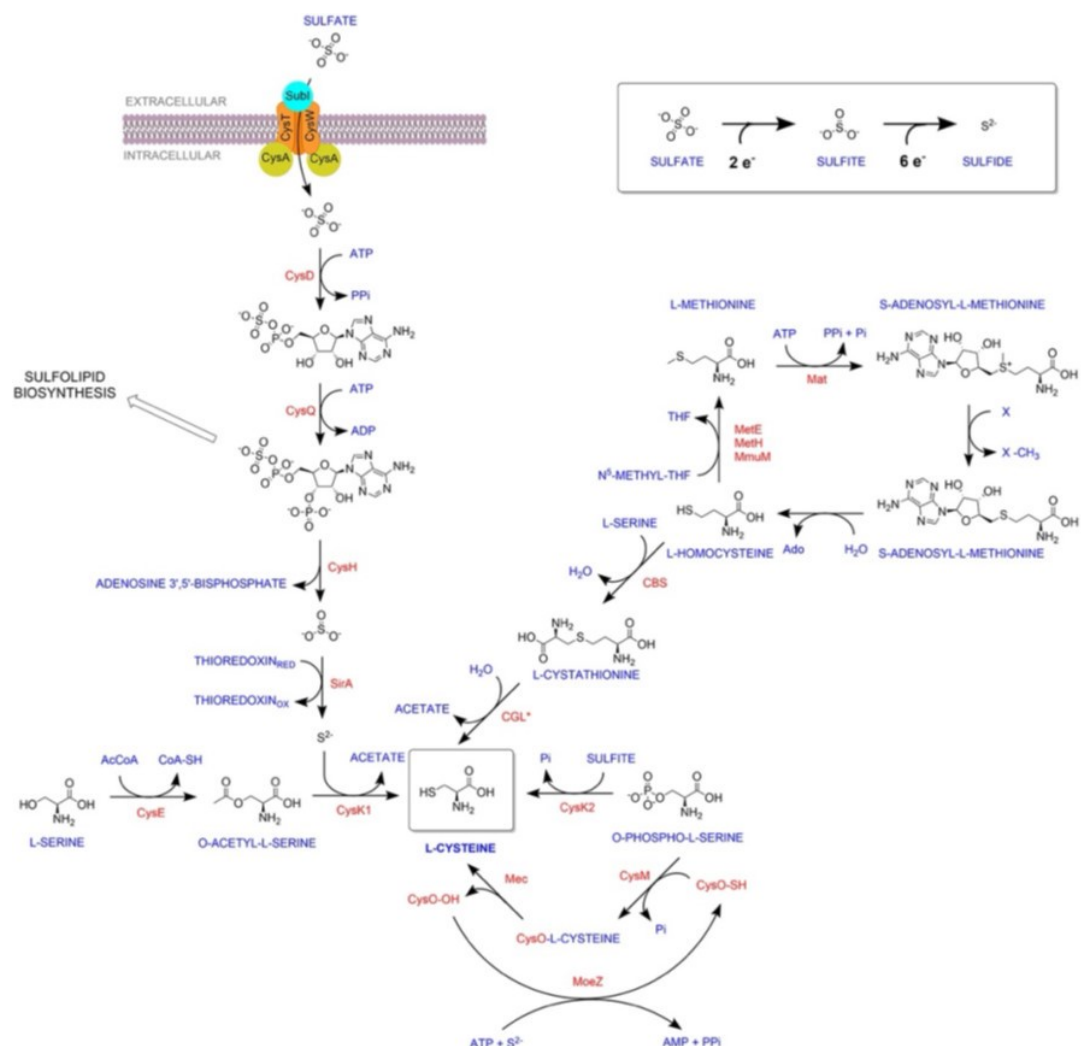
Recommendations For The Authors:

Reviewer #1 (Recommendations For The Authors):

(1) In Figure S1 it would be useful to include the reverse transsulfuration pathway given that it contributes to the L-cysteine pool, and that L-methionine was used for metabolite labelling experiments.

We are in agreement with the reviewer's suggestion, and we have included reverse transsulfuration in Fig S1. Please note that Labelling was not performed with L-methionine. We used ^{34}S derived from SO_4^{2-} to monitor the reductive assimilation of sulfur and its transit from S^{2-} until Lmethionine, passing through cysteine. We specified in material and methods that we have used sodium sulfate- ^{34}S (Merck 718882), as our label source of sulfur. This method was first employed in *M. tuberculosis* by the Bertozzi group to identify sulfolipids in mycobacteria. Therefore, we are not measuring transsulfuration but instead a direct synthesis of Lmethionine via cysteine, and consequently, we are indeed assessing the importance of *cysK2* and *cysM* in this process. We have now added to the results section (page 9) that we employed $(\text{Na}^{34}\text{SO}_4)$ for labeling to make sure other readers will not think we are measuring transsulfuration.

Author response image 1.



(2) In Figure S2 it is unclear why the control is included in this figure given that the stress conditions were compared to the control. What is the control being compared to here?

The heat maps of controls have been included to demonstrate relative gene expression in independent/each of the replicates. The normalized count for the differentially expressed genes are plotted. To better understand the RNA-seq results, we plotted the fold change of differentially expressed genes due to different stress conditions (New figure & table- Figure S3 & Table S2). This allowed us to understand the expression profile of genes in all the stress conditions simultaneously, regardless of whether they were identified as differentially expressed. The data revealed that specific clusters of genes are up- and downregulated in oxidative, SDS, and starvation conditions. In comparison, the differences observed in the pH 5.5 and nitrosative conditions were limited (Figure S3 & Table S2).

(3) In Figure S3 it would be more informative to show fold-enrichment than gene counts in (b) to (f).

In our opinion, gene counts are more informative when plotting GO enrichments, as the number of genes in each GO category can vary drastically. The significance values are already calculated based on the fold enrichment of a category compared to the background, and hence, p-adj values plotted on the x-axis can be sort of a proxy for fold enrichment. Hence, instead of plotting two related variables, plotting the total gene counts that belonged to a category is usually helpful for the reader in understanding the “scale” in which a category is affected.

(4) Figure 1c standard Sautons is a defined media, and is not nutrient-limiting - the authors should clarify the composition of the media that they used here.

The composition of Sautons media used in the study is 0.5g/L $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 2 g/L citric acid, 1g/L L-asparagine, 0.3 g/L $\text{KCl} \cdot \text{H}_2\text{O}$, 0.2% glycerol, 0.64 g/L FeCl_3 , 100 μM NH_4Cl and 0.7 g/L $\text{K}_2\text{HPO}_4 \cdot 3\text{H}_2\text{O}$. We have modified the sentence in line with reviewer’s suggestion.

(5) The authors claim that the distinct transcriptomes for the two mutants indicate that "CysM and CysK2 distinctly modulate 324 and 1104 genes". The effect is likely due to distinct downstream consequences of the deletions, rather than direct regulation by the synthases. This section should be reworded for clarity.

We have modified the sentence in line with reviewer’s suggestion.

(6) In Figure 3 it would be useful to express mycothione levels as a percentage of the total mycothiol pool to give an indication of the extent to which the thiol is being oxidised.

While we appreciate reviewer’s suggestion, we cannot make ratios of IC for two different compounds, as they ionize different. 100 ion counts of one does NOT equal to 100 ion counts of the other.

(7) Figure 6 is difficult to interpret as the concentrations used in the INH + inhibitor wells are not clear. It would be useful to indicate the concentrations of each compound added next to the wells in the figure.

We have modified the figure and legends in line with reviewer’s suggestion

Reviewer #2 (Recommendations For The Authors):

(1) Document the *cysM* deletion.

The details of CysM knockout generation have been previously published ([15]; Appendix Figure S4), and complementation strain details are provided in the methods section.

(2) The oxidative stress CHP is not defined in the figure legend.

We have modified the legend in line with the reviewer's suggestion.

(3) Can we see the structures of the compounds?

Kindly refer to Fig 6a for the structures of compounds

(4) Fix the genetics and the paper is very interesting.

*I might be missing something. The authors do provide promising complementation data for several of the stresses. Provide evidence for the *cysM* deletion and complementation and the data will be very compelling. The focus of the paper is important for our understanding of the biology of *Mycobacterium tuberculosis*.*

Thank you for appreciating our study. The details of CysM knockout and complementation strain generation have been previously published ([15]; Appendix Figure S4 & Methods)). CysK2 mutant and complementation strain details are included in the present manuscript (Figure 1b & Methods).

Reviewer #3 (Recommendations For The Authors):

The transcriptional profiling studies do not rigorously control for the engineered mutations using genetic complementation.

The complementation strains used in all in vitro, ex vivo and in vivo experiments showcase that the phenotypes associated with knockouts are gene specific. We choose not to include complementation strains in RNA sequencing experiments due to the large number of samples handling and associated costs.

Figure 3. These data are not rigorously controlled without genetic complementation, explain why some data in Figure 3 was generated at 24 hr and other data was generated at 48 hr, remove subbars in 3g. Please provide more clarification on Fig 3e-g because the normalization in these panels makes it appear as if there is little- or no-difference in the levels of 34S incorporation into the thiol metabolites.

The complementation strains used in all in vitro, ex vivo, and in vivo experiments showcase that the phenotypes associated with knockouts are gene-specific. We chose not to include complementation strains in Figure 3 experiments due to the large number of sample handling and associated costs.

The time points in the given experiment were chosen based on an initial pilot experiment. It is apparent that a longer duration is required to see the phenotypes associated with labelling compared to pool size. The differences observed are statistically significant.

Surfactant and SDS stress are used interchangeably in the text, legends, and figures. Please be consistent here.

We have modified the text in line with reviewer's suggestion.

Consider re-wording the 1st paragraph on page 5 to better clarify how Trp, Lys, and His interact with the host immune cells.

We have modified the text in line with reviewer's suggestion.

Cite the literature associated with the sulfur import system in Mtb on page 3 in the 2nd paragraph.

We have modified the text in line with reviewer's suggestion.

The manuscript nicely describes the construction of a cysK2 mutant. It is unclear how the cysM mutant was generated. Please clarify, cite, or add the cysM mutant construction to this manuscript.

The details of CysM knockout and complementation strain generation has been previously published ([15]; Appendix Figure S4 & Methods)). We have included the citation in the methods section of current manuscript.

Provide evidence that the small molecules used in Fig 6 are on target and inhibit the cysteine biosynthetic enzymes in whole bacteria. It is unclear how a MIC can be determined with these compounds in 7H9 ADC when deletion mutants grow just fine in this media. Is this because the compounds inhibit multiple cysteine synthesis enzymes and/or enzymatic targets in other pathways? To me, the data suggests that the compounds are hitting multiple enzymes in whole Mtb cells. Does cysteine supplementation reverse the inhibitory profiles with the compounds in Figure 6?

As mentioned in the text, all the compounds were ineffective in killing Mtb, likely because Lcysteine synthases are not essential during regular growth conditions. Hence, the MIC for cysteine inhibitors was very high - C1 (0.6 mg/ml), C2 (0.6 mg/ml), and C3 (0.15 mg/ml) opposed to the standard drug, isoniazid with MIC of 0.06 ug/ml. We agree with the reviewer that the experiments do not conclusively prove that these compounds specifically inhibit the cysteine synthases via "on-target inhibition" in Mtb cells. The inhibitors used in this study have been previously profiled in vitro [8]. However, one cannot rule out the hypothesis that these compounds might also have some off-target effects.

<https://doi.org/10.7554/eLife.91970.2.sa2>